

# MMDIR: Multimodal Instruction-Driven Framework for Mixed-Degradation Document Image Restoration

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## Abstract

Restoring degraded document image is essential for both improving visual quality and optimizing performance in downstream document analysis tasks. Although existing methods have demonstrated substantial improvements in restoration outcomes, they primarily address single-type degradation scenarios. Current approaches typically necessitate training multiple specialized models for specific degradation types or rely on explicit prior knowledge of degradation patterns to guide the training process. To overcome these limitations, we propose *MMDIR*, a multimodal instruction-driven framework designed for document image restoration under mixed and uncertain degradation conditions. By leveraging semantically structured instructions, *MMDIR* dynamically identifies present degradation types (blur, shadow, text watermark, and seal), while enhancing degradation-aware representation learning. Furthermore, we introduce a novel benchmark named *MixedDoc* comprising complex mixed degradations, where each image contains randomized combinations of the aforementioned types. This benchmark addresses a critical gap in existing datasets, which lack realistic multi-degradation samples and often overlook common obstructions such as seals and text watermarks. The effectiveness of our approach is thoroughly validated across both released public benchmarks and our newly proposed dataset. The dataset is available at <https://github.com/xiaomore/MMDIR>.

## 1. Introduction

Document image restoration (DIR) is a crucial preprocessing technology. It is a bridge connecting physical documents with intelligent information processing in the digital world. Its ultimate goal is to maximize the restoration

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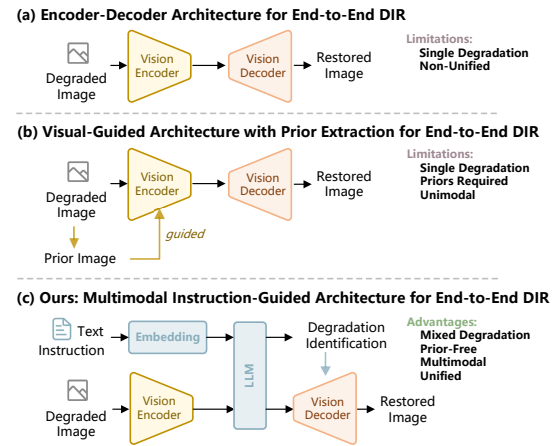


Figure 1. Model patterns for document image restoration (DIR). Existing methods (a) such as DocDiff [40], NAF-DPM [5] and LGA-Doc [34] consist of an end-to-end image-to-image restoration model without any prompt guidance (focusing on single degraded type). The method DocRes [41] (b) uses prior image prompts derived from preprocessed degraded images to guide document restoration, distinguishing between degradation types. Our approach is fundamentally different. As shown in (c), it simultaneously takes multimodal input and output. Text instructions are fed along with the degraded images, prompting the model to answer the instruction question, identifying the presence or absence of degradation types to guide image restoration.

and protection of the information value contained in the document. The advancement of deep learning has accelerated the digitization and intelligent processing of document images. This widespread adoption has significantly enhanced productivity and daily life. However, the development of document intelligence technology inevitably encounters challenges posed by noise-corrupted or degraded image types. Common examples include blurring caused by camera shake during photography, shadows induced by light source positioning and irregular object occlusion, as

well as overlapping seals and text watermarks obscuring main text content in scenarios like contracts and invoices. Therefore, restoring degraded document images is critical for enhancing visual quality, improving the recognizability of key information (text, tables, figures), and supporting downstream tasks such as document understanding, text recognition, table extraction, and image retrieval [9, 23, 39, 42].

Document image restoration (DIR) aims to recover clean, readable document images from degraded inputs, which may suffer from various distortions such as shadows, blur, seals or watermarks. Traditional DIR approaches typically adopt an end-to-end image-to-image framework (as shown in Figure 1 (a)), where a corrupted image is fed into a model (often based on convolutional neural networks (CNN), transformers [35], or generative adversarial networks (GANs) [10]) and supervised using clean reference images to produce a restored output. Representative methods in this category include DocEnTr [33], which leverages vision transformer (ViT) [8] architectures, DE-GAN [32], utilizing adversarial training, and diffusion-based models such as DocDiff [40], NAF-DPM [5] and LGA-Doc [34]. While effective, these methods inherently learn a fixed mapping from degraded inputs to clean outputs without explicit degradation-type conditioning or external guidance. A recent approach, DocRes [41], introduced the idea of incorporating degradation hints during preprocessing. It estimates potential degradation types from the training data and concatenates these weak prompts with the input image to guide the restoration network. Though offering improved control, this strategy necessitates prior knowledge of degradation categories and rigid preprocessing pipelines (illustrated in Figure 1 (b)), failing to generalize to open-world scenarios where degradation types are ambiguous.

The main shortcomings of existing methods are: (i) *Lack of generalization and deviation from real-world scenarios*: Real-world document images often exhibit complex mixed degradations simultaneously (for example, a document may contain shadows, blur, watermarks, and seals concurrently). (ii) *Failure to exploit synergistic effects between tasks*: Certain document restoration tasks may possess intrinsic connections and commonalities. Independently trained models cannot share or utilize this cross-task knowledge during learning, consequently limiting the potential performance ceiling of each individual model.

To bridge this gap, we propose a novel instruction-driven DIR framework that unifies vision and language understanding within a unified cohesive architecture (presented in Figure 1 (c)). Our method accepts degraded image and textual instruction (“*Analyze the image to determine the presence of noise or interference and classify any identified types. And eliminate blur, seal.*”) as inputs. Distinct encoders extract visual and linguistic features, which

are jointly processed by a large language model (LLM). The LLM dynamically interprets the instruction to generate a diagnostic response specifying present/absent degradations and guide a visual decoder toward targeted restoration. The LLM outputs jointly drive a visual decoder to produce the restored image and generate a textual response that answers the user’s query. This multi-task setup enables finer-grained degradation-aware restoration and improved interactivity, bridging the gap between conventional end-to-end DIR and user-guided restoration. This dual-path design enables context-aware processing: the model can identify whether there is a requested removal of degradation in the instruction and detect unmentioned degradations. As illustrated in Figure 1 (c), our approach transcends the limitations of prior art by embedding semantic reasoning into the restoration pipeline, fostering adaptability, transparency.

The main contributions are summarized as follows:

- We propose a instruction-driven multimodal document image restoration (DIR) framework. Our approach integrates vision and language understanding into a unified architecture, enabling more fine-grained, degradation-aware restoration.
- Our model automatically identifies multiple types of degradation in images, guiding restoration and improving the transparency and interpretability. We construct a novel benchmark covering diverse mixed degradation types to systematically address the data gap in this field.
- Experiments demonstrate that our method achieves state-of-the-art performance on both single and mixed degradation scenarios. The code and dataset will be made public to support further research and development.

## 2. Related Work

Document image restoration (DIR) aims to remove unnecessary interference information (watermark, seal, shadow) in the image that is not the original document content, and to repair the original document content that is lost or damaged due to various degradations (such as blur) in the document. In recent years, due to the introduction of Convolutional Neural Networks (CNN), Transformers [35], Generative Adversarial Networks [10] (GAN) and Diffusion model [13], the DIR task has also been widely studied and achieved significant improvements.

**Task-specific Methods.** *Traditional methods* processed the entire degraded image using predefined thresholds [27] or kernel operations [4, 28, 29]. These works were usually effective when the image interference types are relatively simple, but they lack robustness because they based on prior assumptions about the noise patterns in the document.

Existing methods including CNN-based [17, 24], GAN-based [32], Transformer-based [33] and Diffusion-based [5, 34, 40] approaches typically employed an encoder-decoder framework to directly regress noise-free images

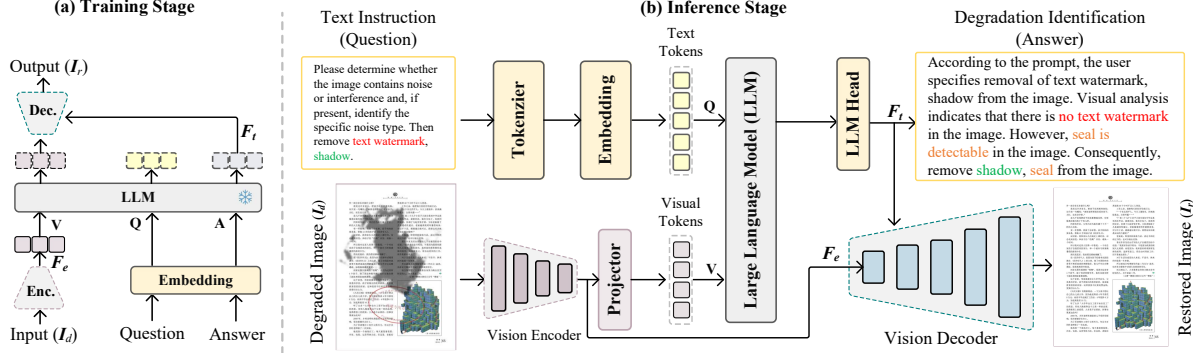


Figure 2. The architecture of our proposed *MMDIR*. Our model handles multimodal inputs and outputs for document image restoration. In training stage as shown in the (a) on the left, it takes a *Degraded Image*  $I_d$ , a corresponding Question (*Text Instruction*) and Answer (*Degradation Identification*) as input. The vision encoder and text embedding first extract multimodal tokens ( $V$ ,  $Q$ ,  $A$ ), which are then fed into a Large Language Model (LLM) to learn cross-modal feature alignment. During the inference stage (in the (b)), only the  $I_d$  and the Question are used as input to the model. In the Question and Answer, degradation types that are not present in the input image are marked in red, while those actually present are highlighted in green. Other types requiring model identification are labeled in orange. LLM generates text responses by decoding step by step to understand the content of instructions and to identify specific degradation types in the input image. The vision decoder incorporates the semantically enriched answer features from the LLM’s output as guidance. It ultimately reconstructs the restored images ( $I_r$ ).

by minimizing discrepancy with ground-truth pixel values. However, they required separate training for different degradation types. Unable to effectively handle multiple and complex mixed degradation scenarios.

**Unified Methods.** DocRes [41] proposed a generalist model to unify multiple document image restoration tasks. It introduced a dynamic task-specific visual prompt that incorporates distinct prior features extracted from the input image to guide the model and provide supplementary information. This highly flexible design enables seamless application to inputs with high and variable resolutions. However, the method relies on the quality of the extracted visual priors, where inaccuracies in feature extraction may directly affect model performance and potentially introduce noise.

For natural scene image restoration, several studies [1, 6, 11, 19, 45] had made significant progress. UniRestore [1] enhanced visual restoration quality by introducing diffusion priors and designing a complementary feature fusion module. DiffUIR [45] proposed a shared distribution term that mapped different degradation types into a unified representation space to guide the restoration. We re-implemented the DiffUIR [45] method using the identical training dataset to enable a comprehensive and fair comparison.

### 3. Proposed Method

In this section, we present our proposed method for document image restoration. As illustrated in Figure 2, the model utilizes an end-to-end multimodal architecture that integrates visual and linguistic understanding. It identifies degradation types in the input image based on textual instructions, where semantic prompts guide the visual decoder to reconstruct a clean image.

#### 3.1. Degradation Identification with Instruction

Existing document image restoration methods typically focus on directly predicting a clean image, without the capability to identify the types of degradation present. Consequently, these models lack the ability to automatically perceive the categories of noise within an image. In previous research on multimodal document understanding, alignment between visual and linguistic modalities [9, 23, 38, 42] has been achieved through tasks such as text recognition or visual question answering (VQA). Given tailored task prompts, these Multimodal Large Language Models (MLLMs) can generate relevant textual responses. Inspired by this, we propose a novel multimodal document image restoration approach, termed Instruction-based Multimodal Document Image Restoration with Degradation Identification. Beyond reconstructing a degradation-free image, our method incorporates a text-modal degradation identification task. This task is designed to establish a spatial alignment between the degradation types described in the text and their corresponding regions in the image, as depicted in Figure 2.

Given a degraded document image  $I_d \in \mathbb{R}^{H \times W \times C}$  as input, where  $H$  and  $W$  is the height and width of the image and  $C = 3$  represents the RGB channels. We generate customized text instruction  $t$  for each input image. The instruction contains two parts: *Identify Degradations* and *Specified Removal* (optional) (e.g., “Please determine whether the image contains noise or interference and, if present, identify the specific noise type. Then remove text watermark, blur, shadow, seal.”).

For visual input, we employ a ViTDet-based [21] architecture as our visual encoder (VisEncoder) to obtain visual tokens. The high-resolution input image  $I_d$  is first split

into non-overlapping patches ( $I_{patches}$ ) with patch size 16. These patches are then processed by multiple transformer layers with windowed self-attention, which effectively captures dependencies among patches. The feature map output by the encoder is recorded as  $F_e$ . Then a Projector containing two  $3 \times 3$  convolutional layers is used to increase the number of channels and reduce the size of the feature map. The resulting visual embeddings are represented as  $V = [v_1, \dots, v_i]$ , where  $i = 256$  denotes the number of visual tokens. For textual input, a tokenizer is applied to both the question (text instruction) and the answer labels (degradation identifications), yielding  $m$  question tokens embedded as  $Q = [q_1, \dots, q_m]$  and  $n$  answer tokens embedded as  $A = [a_1, \dots, a_n]$ . The maximum sequence length for text tokens is set to 128.

Finally, the visual embeddings  $V$  and language embeddings  $Q$  and  $A$  are concatenated along the sequence dimension and fed into the Large Language Model (LLM) shown in Figure 2 (a). This integrated input facilitates cross-modal alignment between vision and language, leading to more accurate degradation type identification.

$$\begin{aligned} F_e &= VisEncoder(I_d) \\ V &= Projector(F_e) \\ F_t &= LLM(CAT(V, Q, A)) \end{aligned} \quad (1)$$

where  $F_t$  is the answer features output by the last layer of LLM.  $CAT$  is concatenation in sequence dimension.

### 3.2. Restored with Semantic Prompts

We leverage the answer features  $F_t$ , which carry textual semantic information, as guiding prompts for image restoration. During inference, these prompts represent a question-guided, speculative semantics rather than a deterministic semantics derived from ground-truth answers. This approach relies on the LLM's commonsense and reasoning capabilities acquired during training, thereby enabling a task-adaptive and semantically-aware image restoration process.

For the vision encoder's output  $F_e$ , we employ an unpatch operation to upsample it into four multi-scale feature maps  $\{F_0, F_1, F_2, F_3\}$ . As illustrated in Figure 3, multi-scale feature fusion is performed following a pyramidal philosophy. The upsampling is achieved using PixelShuffle [31], each time increasing the resolution by a factor of 2. Feature maps at different scales are concatenated along the channel dimension and then fed into a Nonlinear Activation Free Network's (NAF) Block [2]. This process is repeated until the original input image resolution is reconstructed, generating the final restored image ( $I_r$ ). The text tokens  $F_t$  output by LLM is multiplied by a learnable parameter  $W_t$  initialized to 1 as dynamic guided information (*GuidedLayer* in Figure 3) to obtain  $F'_t$ . The visual feature map  $F_e$  is then added to  $F'_t$ , resulting in a text-semantics-guided feature map.

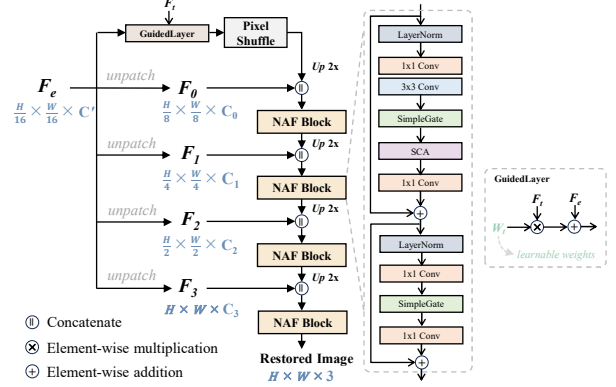


Figure 3. The architecture of vision decoder. *SCA* represents the Simplified Channel Attention in the NAF Block [2]. The *GuidedLayer* weights the feature maps of  $F_e$  and the semantic prompt features  $F_t$  from the LLM. Zoom in for the best view.

$$I_r = VisDecoder(F_e, F_t) \quad (2)$$

During the inference phase, the large language model (LLM) only receives visual tokens ( $V$ ) and text instruction tokens ( $Q$ ) as inputs. It then decodes the Degradation Identification ( $A$ ) step by step, while returning the hidden states at each decoding step and computing the average along the sequence dimension as the hidden states representations ( $F_t$ ) of the predicted answers.

### 3.3. Loss function

During the model training process, we predict degradation identification with Cross-Entropy (CE) loss ( $\mathcal{L}_{ce}$ ). Benefiting from the image-to-image learning paradigm, MMDIR's training objective is to minimize the discrepancy between the restored image and the ground-truth reference. The pixel-level discrepancy is quantified using the  $L_1$  distance between the restored image  $I_r$  and the ground-truth image  $I_{gt}$  ( $\mathcal{L}_{pixel}$ ). Furthermore, a local  $L_1$  loss ( $\mathcal{L}_{local}$ ) is incorporated to enhance the model's perception of specific degradation patterns. We define a binary mask image  $I_{mask}$  (some samples shown in Figure 4) for degradation types such as shadow, watermark, and seal, where the value is 255 in these degradation regions and 0 elsewhere. Additionally, a Structural Similarity Index Measure (SSIM) [37] loss  $\mathcal{L}_{ssim}$  is introduced to evaluate image quality degradation in perceptual space. The formula for the overall training loss is presented below:

$$\begin{aligned} \mathcal{L}_{pixel} &= \|I_{gt} - I_r\|_1 \\ \mathcal{L}_{local} &= \|I_{gt} \times I_{mask} - I_r \times I_{mask}\|_1 \\ \mathcal{L}_{total} &= \mathcal{L}_{pixel} + \alpha \mathcal{L}_{local} + \beta \mathcal{L}_{ssim} + \lambda \mathcal{L}_{ce} \end{aligned} \quad (3)$$

where the balancing factors  $\alpha$ ,  $\beta$  and  $\lambda$  are empirically set to 4, 0.5 and 0.5, respectively.

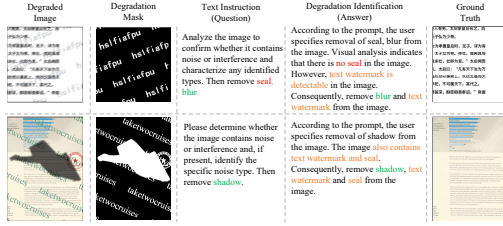


Figure 4. Samples visualization of training dataset.

## 4. Datasets

Our work focuses primarily on restoring document images, especially removing blur, watermarks (specifically textual watermarks), shadows, and seals that frequently occur in real-world scenarios. We categorize the datasets into two types: single degradation and mixed degradation. The *single degradation* set contains images exhibiting only one type of degradation per image, whereas the *mixed degradation* set encompasses images where multiple degradation types co-occur simultaneously.

### Single Degradation Datasets. Document Deblurring.

We utilize the deblurring dataset BMVC [15], which comprises 66k pairs of sharp and blurred document images, each with a size of  $300 \times 300$  pixels. These images originate from diverse document sources and incorporate varied blur kernels to simulate realistic degradation effects. Following the training configurations of DocRes [41], DocDiff [40], NAF-DPM [5], and LGA-Doc[34], we randomly select 30,000 image pairs for training and 10,000 pairs for evaluation.

**Document Deshadow.** Following prior methods, our training set consists of 14,200 synthetic images from FSDSRD [26] and 4,371 real images from the RDD [43] training set. For evaluation, we employ the OSR [36] (237 images) dataset as our test set.

**Mixed Degradation Datasets.** Given the current lack of mixed degradation training and testing datasets, we collect clean and noise-free document images for synthesis. The degradation types in the synthetic dataset include blur, shadow, watermark, and seal.

**Training Dataset.** We collect the clean document images from CDLA [18], CDDOD [20], FSDSRD [26] to synthesize a dataset with mixed degradations including blur, shadow, watermark, or seal. A total of 110k mixed-degradation document images are synthesized for model training, with each image incorporating 1 to 4 randomly combined degradation types (as shown in Figure 4).

For *blur*, we randomly apply one type from Gaussian, Defocus, Glass, Zoom, or Motion Blur, with varied kernel sizes. For *seals*, we systematically generate 20,000 synthetic images by combining random text, dates, shapes, and colors to circumvent the difficulty of real-world data collection. We further augment this set with 1,597 real seal images from DocDiff [40], using 1,277 for training and

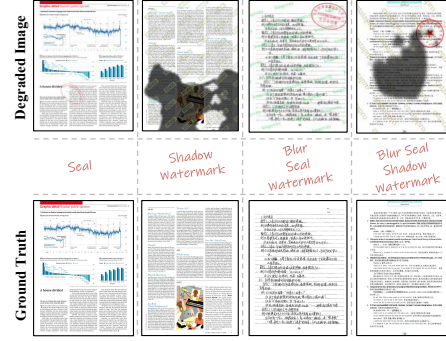


Figure 5. Several samples visualization of our test benchmark MixedDoc. Top row: the degraded images. Middle row: the degradation types *Seal*, *Blur*, *Shadow*, or *Watermark*. Bottom row: the ground-truth images. Zoom in for the best view.

320 for synthesis benchmarks. For *shadows*, we utilize 14,200 masks from FSDSRD [26] and 8,000 from SynShadow [16]. For *text watermarks*, we randomly select phrases or words from our self-collected corpus of Chinese and English text, rendered in diverse colors and fonts under a predefined train-test split. During synthesis, all these elements are randomly cropped and rotated.

**Testing Benchmark.** The source document images are from M6Doc [3]. The synthesis process for the test follows a similar pipeline, with the following distinctions: a separate set of 320 real seal images not used in training is adopted; 432 carefully selected shadow masks from SatelliteWB<sup>1</sup> are used; all elements undergo random cropping and rotation to enhance realism. The final test set, named *MixedDoc*, contains 1,837 images and serves as the evaluation benchmark. Sample images are shown in Figure 5.

## 5. Experiments

**Implementation Details.** We implement the entire model on the PyTorch [30] framework. For the large language model (LLM), we employ Qwen2.5 (0.5B) with its parameters frozen throughout training except the embedding layer. The training image size is set to  $1024 \times 1024$ . We use the AdamW optimizer [25] with a batch size of 16. The learning rate follows a cosine decay schedule with a maximum value of  $3 \times 10^{-4}$ . Our network is trained for 80 epochs with 5% warmup steps on two NVIDIA A100 GPUs.

**Multiple instruction variants and generalizability.** During training, two instruction templates are employed: *no degradation types* specified, *randomly specify the types* (in Table 2). One template and degradation types are randomly chosen for per sample to enhance adaptability. If the specified degradation type *does not exist* in the image, the model will *learn to detect and identify* it during the training phase. During testing, each sample is also evaluated under one of

<sup>1</sup><https://www.kaggle.com/datasets/franciscoescobar/satellite-images-of-water-bodies>

Table 1. Comparisons on single degradation benchmark. “↑” indicates the higher the better and “↓” denotes the opposite. The best performing result is shown in **Bold** font, and the second best result is shown with an underline. The “\*” indicates results reproduced based on the authors’ public code and the same training dataset as used in our method. “256” and “1024” refer to the input image resolutions at low and high settings, respectively.

| Method                    | Venue    | Deblurring (BMVC [15]) |              |               |               | Deshadow (OSR [36]) |              |               |               |
|---------------------------|----------|------------------------|--------------|---------------|---------------|---------------------|--------------|---------------|---------------|
|                           |          | PSNR↑                  | SSIM↑        | LPIPS↓        | DISTS↓        | PSNR↑               | SSIM↑        | LPIPS↓        | DISTS↓        |
| DE-GAN [32]               | TPAMI’19 | 22.30                  | 0.916        | 0.0843        | 0.0968        | -                   | -            | -             | -             |
| DocDiff [40]              | ACMMM’23 | 23.28                  | 0.951        | 0.0307        | 0.0611        | -                   | -            | -             | -             |
| DocShadow [22]            | ICCV’23  | -                      | -            | -             | -             | 18.25               | 0.902        | -             | -             |
| BGSNet [43]               | CVPR’23  | -                      | -            | -             | -             | <b>21.25</b>        | <b>0.931</b> | 0.0702        | 0.0995        |
| NAF-DPM [5]               | arXiv’24 | 27.61                  | 0.975        | 0.0207        | <u>0.0409</u> | -                   | -            | -             | -             |
| DocRes [41]               | CVPR’24  | 27.35                  | 0.972        | -             | -             | 17.37               | 0.923        | 0.0664        | 0.1035        |
| LGA-Doc [34]              | ICMR’25  | <u>28.95</u>           | <b>0.978</b> | <u>0.0169</u> | 0.0470        | -                   | -            | -             | -             |
| DocDiff <sub>256</sub> *  | -        | 25.19                  | 0.953        | 0.0257        | 0.0471        | 18.06               | 0.785        | 0.2806        | 0.2176        |
| DocDiff <sub>1024</sub> * | -        | 24.95                  | 0.952        | 0.0261        | 0.0477        | 18.80               | 0.912        | 0.0703        | 0.1209        |
| DiffUIR <sub>256</sub> *  | -        | 22.68                  | 0.910        | 0.0618        | 0.1000        | 18.47               | 0.796        | 0.2724        | 0.2041        |
| DiffUIR <sub>1024</sub> * | -        | 23.09                  | 0.914        | 0.0557        | 0.0752        | <u>21.08</u>        | <u>0.928</u> | <u>0.0579</u> | <u>0.0848</u> |
| MMDIR (Ours)              | -        | <b>29.03</b>           | <u>0.977</u> | <b>0.0150</b> | <b>0.0233</b> | 19.60               | <u>0.928</u> | <b>0.0547</b> | <b>0.0816</b> |

Table 2. Instruction variants.

| Type               | Instruction Template Example                                                                                                                                        |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No specified       | Evaluate the image for the presence of noise or interference and specify the type(s) detected.                                                                      |
| Randomly specified | Please determine whether the image contains noise or interference and, if present, identify the specific noise type. Then remove <i>RANDOM SELECTED 1→4 TYPES</i> . |

two variants like training stage. Results are averaged across samples to evaluate model stability and generalization.

**Evaluation metrics.** We follow previous methods [22, 34, 40, 41], by combining pixel-level comparison metrics with perceptual similarity metrics. The pixel-level metrics include Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) [14]. **PSNR** quantifies image quality by calculating the signal-to-noise ratio, while **SSIM** evaluates structural similarity between the distortion-free and distorted images in terms of luminance, contrast, and structure. Perceptual similarity metrics include Learned Perceptual Image Patch Similarity (LPIPS) [44] and Deep Image Structure and Texture Similarity (DISTS) [7]. **LPIPS** employs deep learning to better reflect perceptual differences compared to traditional metrics. **DISTS** further improves evaluation by incorporating structural and texture sensitivity through a deep neural network, thereby providing a more comprehensive quality assessment.

### 5.1. Results on single degradation benchmarks

**Quantitative Results.** In Table 1, we compare our method with task-specific works and unified frameworks (DocRes [41], DiffUIR [45]). DocRes [41] is a unified framework but lacked the prior extraction for seals and watermarks. DocDiff [40] performed seal and watermark removal, yet its training data was not released. DiffUIR [45] is a recent and excellent unified natural scene image restoration method based on diffusion model, we transfer it to the document

Table 3. Comparisons on our proposed mixed degradation benchmark (MixedDoc).

| Method                    | PSNR↑        | SSIM↑        | LPIPS↓        | DISTS↓        |
|---------------------------|--------------|--------------|---------------|---------------|
| DocDiff <sub>256</sub> *  | 18.35        | 0.744        | 0.4392        | 0.3082        |
| DocDiff <sub>1024</sub> * | 21.14        | 0.863        | 0.1923        | 0.1966        |
| DiffUIR <sub>256</sub> *  | 18.79        | 0.742        | 0.4525        | 0.3093        |
| DiffUIR <sub>1024</sub> * | 22.09        | 0.853        | 0.2008        | 0.2064        |
| MMDIR (Ours)              | <b>24.43</b> | <b>0.908</b> | <b>0.1217</b> | <b>0.1323</b> |

domain. To ensure a consistent training set and the fair comparison, we retrain DocDiff and DiffUIR using our training data, marking these results with “\*” in Table 1 and 3.

The results in Table 1 demonstrate that our method surpasses prior works on most metrics. Regarding *deblurring*, our approach yields competitive results: it attains SOTA on three metrics, with only the SSIM score being slightly lower (by 0.001) compared to LGA-Doc. Most importantly, we observe significant improvements of 11.2% on LPIPS and 43.0% on DISTS. These results suggest that our model successfully recovers image edges and textural information, generating outputs that are not only more natural and authentic but also perceptually more aligned with the ground-truth images.

For *deshadow* on OSR benchmark [36], our method also demonstrates a significant advantage in perceptual quality, achieving the best performance on both LPIPS and DISTS metrics. This observation, consistent with the findings in deblurring, collectively validates the effectiveness of our proposed unified framework in restoring visual realism and structural integrity. While methods like BGSNet [43] and DiffUIR [45] achieved higher scores in PSNR or SSIM, they often produce over-smoothed or blurry outputs, as evidenced by their poorer LPIPS and DISTS results. Therefore, these results confirm that the optimization direction of our method aligns more closely with human visual perception and reflects its strong capability in generating realistic details and textures.

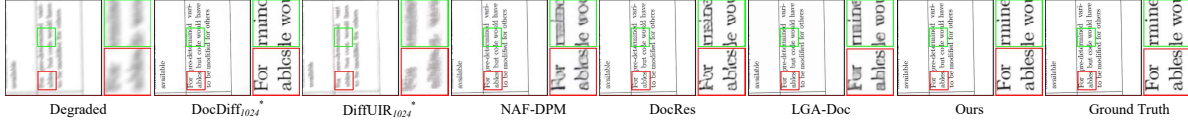


Figure 6. Visual Comparison on the deblurring benchmark BMVC [15]. The first column “Degraded” is input image, and the last column “Ground Truth” indicates clean reference. Red and green boxes highlight zoomed-in detail discrepancies.

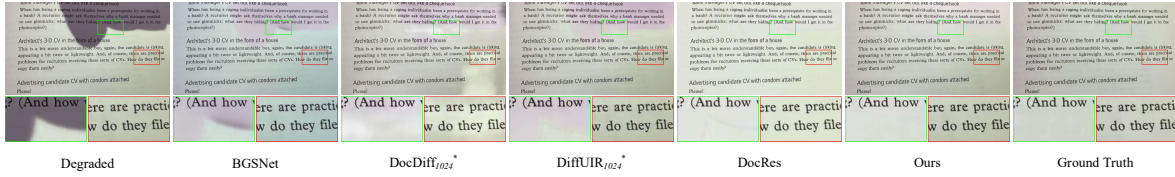


Figure 7. Visual Comparison on the deshadow Benchmark OSR [36].

**Qualitative Analysis.** To intuitively evaluate the performance, we conduct a visual comparison to further validate the effectiveness of our proposed method, as shown in Figure 6 and 7. We compare our approach against existing prominent methods. The visual results demonstrate that, in the global restoration quality, our method achieves blur and shadow removal results closer to the ground-truth image, producing visually clearer character strokes (in Figure 6 for deblurring) and more realistic outputs. Furthermore, in the local detail areas marked by the red and green boxes, our method more accurately distinguishes shadow regions from background content at shadow boundaries (in Figure 7). This effectively mitigates issues of residual shadows and over-erasure, showcasing its superior capabilities in detail preservation and boundary recovery.

## 5.2. Results on mixed degradation benchmark

**Quantitative Results.** We conduct experiments on our proposed mixed degradation benchmark (MixedDoc), where each image contains 1 to 4 randomly combined degradation types. In Table 3, our method comprehensively surpasses existing methods like DocDiff and DiffUIR across all four metrics at both 256 and 1024 resolutions. This result convincingly demonstrates the superior generalization and robustness of our model in handling uncertain and mixed image degradations. Compared to other methods, our model more effectively restores multiple co-existing degradations, thereby producing high-quality, high-fidelity results even under highly complex degradation conditions.

**Qualitative Analysis.** As shown in Figure 8, we present image restoration examples covering a variety of typical scenarios, including scanned images of magazines and books, as well as real-world images captured by mobile devices. We compare our method with re-implemented versions of DocDiff and DiffUIR trained on the same dataset. Experimental results demonstrate that our approach consistently achieves superior restoration performance, both on single and mixed degradation types. In terms of local texture and structural recovery, the proposed method gener-

Table 4. Ablation studies of text instruction on Deblurring (BMVC [15]), Deshadow (OSR [36]) and our proposed benchmark Mixed-Doc. *Inst.* means text instruction. “✓” indicates that the instruction is used. “✗” is the opposite.

| Bench.   | Inst. | PSNR↑/SSIM↑        | LPIPS↓/DISTS↓        |
|----------|-------|--------------------|----------------------|
| BMVC[15] | ✗     | 26.76/0.966        | 0.0219/0.0320        |
|          | ✓     | <b>29.03/0.977</b> | <b>0.0150/0.0233</b> |
| OSR[36]  | ✗     | 18.79/0.922        | 0.0610/0.0960        |
|          | ✓     | <b>19.60/0.928</b> | <b>0.0547/0.0816</b> |
| MixedDoc | ✗     | 23.83/0.898        | 0.1324/0.1489        |
|          | ✓     | <b>24.43/0.908</b> | <b>0.1217/0.1323</b> |

ates more realistic and natural details. Notably, even in highly complex degradation scenarios as shown in the last example, our method still exhibits stronger robustness and restoration capability.

Figure 9 provides visual examples of our model’s output with multi-modal inputs. As shown, the model can accurately identify the actual degradation types present in the input degraded image based on the text instruction. Additionally, the heatmaps (Attn in Figure 9) show that our method effectively focuses on the relevant regions within the document.

**FID, OCR Performance and Computational Cost.** As shown in Table 5. FID [12] is Fréchet Inception Distance. Character Error Rate (CER) and Edit Distance (ED) are OCR evaluation metrics. The open-source PaddleOCR serves as the recognition engine. We **retrained FoundIR** [19], and tested on the MixedDoc benchmark. The results are PSNR/22.33, SSIM/0.86, LPIPS/0.195, DISTS/0.198. Our method demonstrates superiority in comparison.

Table 6 and 7 present the performance of per-type and varying degradation numbers on MixedDoc benchmark. The recall declines as the number of degradation types increases, while precision fluctuates—a pattern consistent with multi-label classification. This reflects conservative predictions on complex samples (higher precision, lower recall) and more relaxed predictions on simpler ones.

**Ablation on Text Instructions.** To verify the effective-

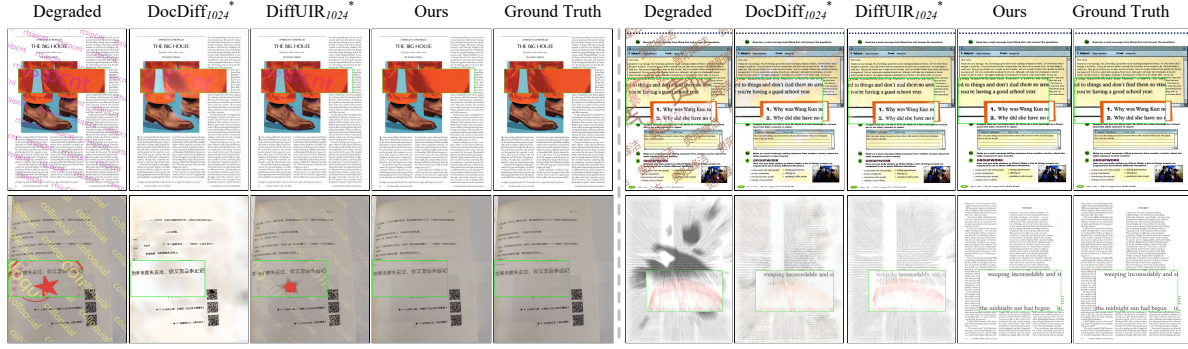


Figure 8. Qualitative visual on our proposed MixedDoc benchmark. The “\*” denotes results from retraining previous methods (DocDiff [40] and DiffUIR [45]) on the same training set, while the green zoomed-in boxes highlight detailed differences.

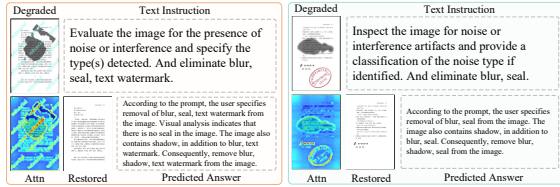


Figure 9. Visualization of our model’s predictions.

Table 5. OCR (CER/ED), FID, model parameters and inference speed results on MixedDoc benchmark. The input image size is  $1024^2$  for all methods. The speed is influenced by input size, model architecture, and diffusion model sampling time steps ( $T$ ).

| Method   | CER↓        | ED↓          | FID↓        | $T$ | Speed        | #Param. |
|----------|-------------|--------------|-------------|-----|--------------|---------|
| DocDiff* | 0.27        | 368.2        | 24.62       | 100 | 3.02s        | 8M      |
| DiffUIR* | 0.24        | 352.9        | 21.45       | 2   | 1.80s        | 36M     |
| FoundIR* | 0.23        | 328.5        | 10.41       | 4   | 3.54s        | 36M     |
| Ours     | <b>0.17</b> | <b>240.8</b> | <b>9.07</b> | -   | <b>1.51s</b> | 735M    |

Table 6. Precision (P), Recall (R) and F1 of per degradation type on MixedDoc Benchmark.

| Degradation Type | P     | R     | F1    |
|------------------|-------|-------|-------|
| blur             | .9772 | .9407 | .9586 |
| seal             | .9812 | .9732 | .9772 |
| shadow           | .9985 | .9638 | .9808 |
| textWatermark    | .9993 | .9985 | .9989 |

Table 7. Precision (P), Recall (R), and F1 for different numbers of degradations on MixedDoc Benchmark.

| Mixed | Samples | Percentage | P     | R     | F1    |
|-------|---------|------------|-------|-------|-------|
| 1     | 110     | 6.0%       | .9910 | 1     | .9955 |
| 2     | 693     | 37.7%      | .9826 | .9784 | .9805 |
| 3     | 786     | 42.8%      | .9913 | .9712 | .9811 |
| 4     | 248     | 13.5%      | 1     | .9647 | .9820 |

ness of text instructions, we design corresponding ablation experiments. Without incorporating text guidance (without using the Large Language Model), the model structure retains only the vision encoder and decoder. Quantitative evaluation results on three benchmark datasets are shown in Table 4. Experimental data demonstrate that introducing text instructions comprehensively and significantly im-

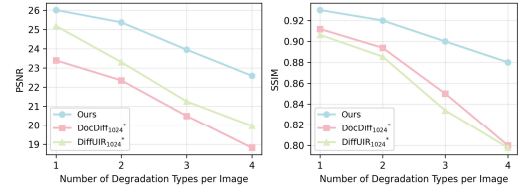


Figure 10. Trend of PSNR and SSIM with increasing degradation types on the MixedDoc benchmark.

proves all performance metrics for document restoration.

**Impact of the Number of Degradation Types.** We analyze the restoration results under mixed degradation scenarios. As shown in Figure 10, when the number of degradation types per image increases from one to four, both PSNR and SSIM metrics demonstrate a declining trend. This phenomenon is highly intuitive. We attribute it primarily to the complexity and interactions of the degradations. Each degradation type has unique statistical properties, such as its distribution, intensity, and spatial correlation. When multiple degradations coexist, they interact and combine into more complex noise patterns, corrupting information across frequency bands. As a result, when the model attempts to remove all noises, it risks over-smoothing or mistakenly eliminating crucial details like text edges.

## 6. Conclusion

In this paper, we propose MMDIR, a multimodal document image restoration method guided by text instructions, to address mixed degradations in real scenarios. Unlike conventional works that limited to single degradation types and reliance on specific prior knowledge, MMDIR bridges visual and linguistic modalities to achieve semantic-aware mixed degradation restoration. We also introduce MixedDoc, a benchmark covering diverse mixed degradations. Experiments show state-of-the-art results on both public benchmarks and MixedDoc, demonstrating superior robustness in single and mixed degradation scenarios. Future work will expand degradation types and enhance data authenticity to improve document restoration.

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